

Generative Modelling for Realistic Mars Landing Site Visualisation: Methodology

Deepakraj Rajmohan
MSc in Artificial Intelligence
University of Surrey
deepak.dr036@gmail.com

Abstract—Photorealistic virtual-reality training for Mars rover operators demands surface imagery at the centimetre-scale resolution that rover cameras deliver, yet no source of paired nadir-to-perspective Mars imagery exists. This methodology chapter presents a three-stage physics-constrained generative pipeline that closes that gap without requiring geographically aligned training pairs. Stage 1 optionally super-resolves CaSSIS or CTX orbital imagery to HiRISE-equivalent resolution (25 cm/pixel) via diffusion-based super-resolution [1], [2]. Stage 2 applies an Unpaired Neural Schrödinger Bridge [3] to translate HiRISE nadir patches into synthetic rover-perspective images; four physics-informed loss terms enforce solar-geometry consistency, Hapke bidirectional reflectance [4], geological morphometry, and perceptual realism. Stage 3 trains a 3D Gaussian Splatting scene [5] on the synthetic multi-view set to yield real-time VR-compatible renders. The chapter details the mathematical formulation, dataset construction (80,000 images total), training strategy, and an evaluation protocol benchmarked against four baselines on metrics spanning perceptual quality, physical plausibility, and hallucination rate.

I. RESEARCH OBJECTIVES AND SCOPE

The overarching aim of this dissertation is to develop and evaluate a generative pipeline that synthesises photorealistic rover-perspective surface imagery of Mars from existing orbital remote-sensing products, and to embed that imagery in a real-time virtual-reality environment suitable for mission training. The literature review established that no existing method closes the nadir-to-rover-perspective gap for Mars surface synthesis: single-image super-resolution methods are limited to $4\times$ upsampling [6], [7]; unpaired domain-translation approaches such as CycleGAN lack physics grounding and produce systematic photometric errors; and Mars-specific neural radiance field reconstructions to date rely on rover imagery that is unavailable prior to landing [8]. The proposed system must therefore satisfy the following objectives.

- O1.** Produce nadir imagery at 25 cm/pixel from coarser CaSSIS/CTX inputs when HiRISE coverage is absent.
- O2.** Translate unpaired HiRISE nadir patches into statistically plausible rover-perspective images whose photometric and morphometric properties are consistent with Mars surface physics.
- O3.** Reconstruct a real-time renderable 3D scene from the synthetic perspective images at VR frame rates (≥ 72 fps).

- O4.** Quantify the fidelity and physical plausibility of generated imagery against held-out real rover data and known planetary surface statistics.

The scope is restricted to visible-wavelength monochromatic and RGB imagery. Thermal infrared, radar, and spectral modalities are excluded. The primary test site is the ExoMars 2028 candidate landing area, Oxia Planum (18.3° N, 335.5° E), selected because its clay-bearing terrain and geomorphic diversity make it a challenging but well-characterised benchmark. Secondary validation uses geologically similar terrain patches drawn from the HiRISE archive to assess generalisability.

II. PROBLEM FORMULATION

A. Notation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an image tensor with spatial dimensions $H \times W$ and C channels ($C = 1$ for grayscale, $C = 3$ for RGB). Define two domains: the nadir orbital domain $\mathcal{X} = \{\mathbf{x}_i^{\text{nadir}}\}_{i=1}^N$ comprising HiRISE patches at 25 cm/pixel, and the rover-perspective domain $\mathcal{Y} = \{\mathbf{y}_j^{\text{rover}}\}_{j=1}^M$ comprising NAVCAM images. No correspondence between index sets i and j exists; the training regime is fully unpaired. Let P_{nadir} and P_{persp} denote the respective marginal distributions on $\mathbb{R}^{512 \times 512 \times 1}$.

B. Stage 1: Diffusion-Based Super-Resolution

Given a low-resolution CaSSIS or CTX image $\mathbf{x}^{\text{lr}}$ at resolution $r_{\text{in}} \in \{4, 6\}$ m/pixel, Stage 1 produces an estimate $\hat{\mathbf{x}}^{\text{hr}}$ at $r_{\text{out}} = 0.25$ m/pixel, corresponding to scale factors of $16\times$ and $24\times$ respectively. Following the conditional diffusion framework of DiffusionSat [1] and the taming-diffusion super-resolution approach [2], the model learns a denoising score function $s_\theta(\mathbf{z}_t, t, \mathbf{x}^{\text{lr}})$ conditioned on the low-resolution image, where $\mathbf{z}_t$ is the noisy latent at diffusion time $t \in [0, T]$. The reverse SDE is:

$$d\mathbf{z}_t = [f(\mathbf{z}_t, t) - g(t)^2 s_\theta(\mathbf{z}_t, t, \mathbf{x}^{\text{lr}})] dt + g(t) d\bar{\mathbf{W}}_t, \quad (1)$$

where f and g are the drift and diffusion coefficients of the VP-SDE schedule, and $\bar{\mathbf{W}}_t$ is a reverse-time Wiener process. Stage 1 is optional when HiRISE imagery is available directly from the NASA Planetary Data System (PDS) archive.

C. Stage 2: Neural Schrödinger Bridge for View Translation

The core contribution is a stochastic transport map from P_{nad} to P_{persp} that requires no paired supervision. The Neural Schrödinger Bridge (UNSB) [3] solves the entropy-regularised optimal transport problem:

$$\pi^* = \arg \min_{\pi \in \Pi(P_{\text{nad}}, P_{\text{persp}})} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \pi} [c(\mathbf{x}, \mathbf{y})] + \varepsilon \text{KL}(\pi \| R), \quad (2)$$

where $c(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2$ is the squared ℓ_2 transport cost, $\varepsilon > 0$ is an entropic regularisation coefficient (set to $\varepsilon = 0.05$ in our experiments), and R is the reference Wiener measure. The optimal coupling π^* is realised by a forward-backward SDE pair. The forward SDE transports samples from P_{nad} toward P_{persp} :

$$d\mathbf{x}_t = \underbrace{f_\phi(\mathbf{x}_t, t)}_{\text{drift net}} dt + \sigma d\mathbf{W}_t, \quad t \in [0, 1], \quad (3)$$

where f_ϕ is a learned drift parameterised by a U-Net with attention [3], σ controls the diffusion coefficient, and $\mathbf{x}_0 \sim P_{\text{nad}}$, $\mathbf{x}_1 \approx P_{\text{persp}}$. The reciprocal process provides the backward SDE, and the two processes are jointly trained via the IPFP (Iterative Proportional Fitting Procedure) variant described in [3]. For applications requiring deterministic outputs (e.g. repeated rendering from the same nadir patch), the SDE can be replaced by the Brownian Bridge ODE of Xiao et al. [9], eliminating stochasticity entirely.

D. Physics-Informed Loss Functions

Unconstrained image translation networks trained on Mars data are prone to photometric hallucinations [10], [11]. We introduce four physics-informed regularisation terms appended to the standard Schrödinger Bridge loss $\mathcal{L}_{\text{SB}}$:

a) Solar geometry consistency ($\mathcal{L}_{\text{solar}}$): Let (α_s, β_s) denote the sun azimuth and elevation angle at the HiRISE acquisition epoch, extracted from image metadata. A shadow-direction estimator D_ψ predicts the dominant shadow direction $\hat{\theta}_{\text{shad}}$ from the generated image $\hat{\mathbf{y}} = G_\phi(\mathbf{x})$. The expected shadow direction θ_s^* follows from solar geometry as $\theta_s^* = (\alpha_s + 180^\circ) \bmod 360^\circ$. The loss penalises angular deviation:

$$\mathcal{L}_{\text{solar}} = \mathbb{E}[\|\hat{\theta}_{\text{shad}} - \theta_s^*\|_\angle], \quad (4)$$

where $\|\cdot\|_\angle$ denotes the smallest circular angle distance in degrees.

b) Hapke photometric constraint ($\mathcal{L}_{\text{photo}}$): The Hapke bidirectional reflectance distribution function (BRDF) [4] models Mars surface reflectance as:

$$r(\mu_0, \mu, g) = \frac{\varpi_0}{4(\mu_0 + \mu)} \times [(1 + B_0 B(g))P(g) + H(\mu_0)H(\mu) - 1], \quad (5)$$

where $\mu_0 = \cos(\theta_i)$ and $\mu = \cos(\theta_e)$ are the cosines of the incidence and emission angles, g is the phase angle, $\varpi_0 \approx 0.22$ is the single-scattering albedo for Mars ferric oxide surfaces, $B(g)$ is the opposition surge function, $P(g)$ is the single-particle phase function, and $H(\cdot)$ is the Ambartsumian–Chandrasekhar function. A pretrained Hapke inversion network Ω_ξ estimates ϖ_0 from a generated patch. The photometric

loss minimises deviation from the prior albedo distribution $\mathcal{N}(\mu_\varpi, \sigma_\varpi^2)$ with $\mu_\varpi = 0.22$ and $\sigma_\varpi = 0.05$:

$$\mathcal{L}_{\text{photo}} = \mathbb{E} \left[\frac{(\Omega_\xi(\hat{\mathbf{y}}) - \mu_\varpi)^2}{\sigma_\varpi^2} \right]. \quad (6)$$

c) Geological morphometry prior ($\mathcal{L}_{\text{geo}}$): Mars surface patches at rover scale contain diagnostic morphological signatures: impact craters follow a power-law cumulative size-frequency distribution (SFD) $N(> D) \propto D^{-b}$ with $b \approx 2$ for primary craters [12], and aeolian bedform wavelength-to-height ratios cluster near 20–40 at Oxia Planum elevations. A pretrained morphometric classifier K_κ outputs a scalar in-distribution score $s_{\text{geo}} \in [0, 1]$, where $s_{\text{geo}} = 1$ indicates the patch statistics are consistent with the known Mars prior. The geological loss is:

$$\mathcal{L}_{\text{geo}} = \mathbb{E}[1 - s_{\text{geo}}(K_\kappa(\hat{\mathbf{y}}))]. \quad (7)$$

d) Perceptual realism ($\mathcal{L}_{\text{perceptual}}$): Following standard practice in image translation, a VGG-19-based perceptual loss [13] compares activations at layers `relu2_2` and `relu3_3`:

$$\mathcal{L}_{\text{perceptual}} = \sum_{l \in \{2,3\}} \mathbb{E}[\|\phi_l(\hat{\mathbf{y}}) - \phi_l(\mathbf{y}_{\text{ref}})\|_1], \quad (8)$$

where $\mathbf{y}_{\text{ref}}$ is a randomly sampled real rover image (no geographic correspondence assumed) and ϕ_l extracts layer- l features.

e) Total training objective: The full loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SB}} + \lambda_1 \mathcal{L}_{\text{solar}} + \lambda_2 \mathcal{L}_{\text{photo}} + \lambda_3 \mathcal{L}_{\text{geo}} + \lambda_4 \mathcal{L}_{\text{perceptual}}, \quad (9)$$

with hyperparameters $\lambda_1 = 0.5$, $\lambda_2 = 1.0$, $\lambda_3 = 0.8$, and $\lambda_4 = 0.1$. These values are set on the basis of preliminary scale analysis: $\mathcal{L}_{\text{solar}}$ ranges over $[0^\circ, 180^\circ]$ and is divided by 180 before weighting; $\mathcal{L}_{\text{photo}}$ is dimensionless; $\mathcal{L}_{\text{geo}}$ is bounded in $[0, 1]$; and $\mathcal{L}_{\text{perceptual}}$ is ℓ_1 feature distance, which empirically runs one order of magnitude larger. Final λ values will be refined via grid search on the validation set.

E. Stage 3: Gaussian Splatting Scene Reconstruction

Given a set of K synthetic perspective images $\{\hat{\mathbf{y}}_k, \mathbf{P}_k\}_{k=1}^K$, where $\mathbf{P}_k \in SE(3)$ is the estimated camera pose, Stage 3 fits a 3D Gaussian Splatting (3DGS) scene representation [5]. Each Gaussian is parameterised by a mean position $\boldsymbol{\mu}_j \in \mathbb{R}^3$, a covariance matrix $\boldsymbol{\Sigma}_j$ encoded via quaternion and scale, an opacity $\alpha_j \in [0, 1]$, and a spherical harmonic colour vector $\mathbf{c}_j$. The scene is optimised by minimising the photometric reconstruction loss:

$$\mathcal{L}_{\text{3DGS}} = \sum_{k=1}^K \left[(1 - \lambda_{\text{dssim}}) \|\mathcal{R}(\mathcal{G}, \mathbf{P}_k) - \hat{\mathbf{y}}_k\|_1 + \lambda_{\text{dssim}} \mathcal{L}_{\text{D-SSIM}}(\mathcal{R}(\mathcal{G}, \mathbf{P}_k), \hat{\mathbf{y}}_k) \right], \quad (10)$$

where $\mathcal{R}(\mathcal{G}, \mathbf{P}_k)$ is the differentiable rasteriser producing the rendered view, and $\lambda_{\text{dssim}} = 0.2$ following [5]. Pose estimates $\mathbf{P}_k$ are obtained by assigning synthetic viewpoints based on the known DTM geometry of the target area combined with the perspective parameters used in Stage 2 image generation.

[Figure placeholder: Three-stage pipeline diagram. Left block — Stage 1 (Super-Resolution): CaSSIS/CTX orbital image at 4–6 m/pixel enters a conditional diffusion model; the output is a 25 cm/pixel HiRISE-equivalent nadir patch. A dashed bypass arrow labelled “HiRISE available” skips Stage 1. Centre block — Stage 2 (Neural Schrödinger Bridge): the HiRISE patch and solar-geometry metadata enter the UNSB translation network. Four coloured loss branches (solar; Hapke, geological, perceptual) feed back into the network during training. The output is a batch of synthetic rover-perspective images. Right block — Stage 3 (3D Gaussian Splatting): the synthetic multi-view set and estimated camera poses are input to 3DGS optimisation; the output is a real-time renderable scene displayed in a VR headset. Arrows between blocks are labelled with tensor shape annotations.]

Fig. 1. Overview of the three-stage physics-constrained generative pipeline. Stage 1 (left) optionally super-resolves coarse orbital imagery to HiRISE-equivalent 25 cm/pixel resolution using a conditional diffusion model; this stage is bypassed when HiRISE data are already available. Stage 2 (centre) applies the Unpaired Neural Schrödinger Bridge [3] to transport nadir image patches from the orbital distribution P_{nadir} to the rover-perspective distribution P_{persp} without requiring geographically aligned training pairs; four physics-informed loss branches (solar geometry, Hapke BRDF [4], geological morphometry, and perceptual realism) constrain the generated imagery to lie within physically plausible bounds. Stage 3 (right) fits a 3D Gaussian Splatting scene [5] to the synthetic perspective image set and renders it at real-time VR frame rates. Dashed arrows indicate optional flows; solid arrows indicate mandatory data flow.

III. PROPOSED SYSTEM ARCHITECTURE

The three stages of the pipeline are designed to be modular: Stages 2 and 3 can operate independently of Stage 1 whenever HiRISE coverage is available, which covers approximately 3% of the Martian surface at the time of writing.

A. Stage 1 — Nadir Super-Resolution

Stage 1 addresses the resolution gap between CaSSIS (4 m/pixel) or CTX (6 m/pixel) and HiRISE (25 cm/pixel). Upsampling by factors of 16–24 lies far beyond the range of conventional interpolation or standard $4\times$ CNN methods [6]. A latent diffusion model conditioned on the low-resolution input [1], [2] generates multiple plausible high-resolution realisations; the median of five samples is used as the Stage 2 input to reduce single-sample noise. The model architecture follows the VQ-VAE + conditional U-Net design of DiffusionSat [1], fine-tuned on paired (CTX, HiRISE) image crops from Oxia Planum and geologically diverse Martian terrain drawn from the MCTED dataset [14]. Where paired orbital data are available at both resolutions over the same footprint, supervised training is straightforward; where only single-resolution data exist, the model is trained with synthetic degradation kernels.

B. Stage 2 — Physics-Constrained View Translation

Stage 2 is the central contribution of this dissertation. The UNSB framework [3] is chosen over CycleGAN and standard diffusion inpainting [13] for three reasons. First, it minimises an entropy-regularised optimal transport objective (Eq. 2), providing a principled probabilistic interpretation of the domain gap. Second, it supports unpaired training, which is essential because no geographically registered nadir–perspective pairs exist for Mars. Third, the SDE formulation (Eq. 3) permits controllable stochasticity: switching to a Brownian Bridge ODE [9] yields deterministic outputs from the same nadir patch,

which is useful for systematic evaluation and for generating multi-view consistency in Stage 3.

The generator G_ϕ is a 256-channel U-Net with eight residual blocks and self-attention at 32×32 and 16×16 spatial scales. A patch-based discriminator D_ψ is used solely in the perceptual loss branch; it is not adversarially trained in the conventional GAN sense because the SB objective itself provides the mode-covering guarantee. Solar azimuth and elevation metadata are encoded as a two-element sinusoidal embedding and injected at each residual block via adaptive layer normalisation, following the conditioning strategy of PFMGAN [15]. This allows the model to produce shadow orientations that are consistent with the acquisition geometry of the input nadir patch.

At inference, a single HiRISE patch of size 512×512 pixels is processed in approximately 0.4 s on a single A100 GPU. The stage outputs a set of K synthetic rover-perspective images generated from perturbed viewpoint parameters; $K = 24$ images per patch are used in Stage 3 experiments.

C. Stage 3 — Gaussian Splatting VR Rendering

Stage 3 trains a 3DGS scene [5] on the K synthetic images produced by Stage 2. Camera intrinsics are set to match the Curiosity MastCam-Z field of view (52° horizontal), and extrinsics are placed on a hemisphere of radius 1.5 m centred at the terrain patch midpoint, sampled at 15° azimuth intervals and two elevations (15° and 45° above horizontal) to mimic rover viewing geometry. 3DGS optimisation converges in approximately 30,000 iterations (roughly 35 min on a single A100). The resulting scene renders at >1500 fps at 1920×1080 resolution in the absence of VR reprojection overhead, comfortably exceeding the 72 fps requirement of commodity VR headsets. Surface normals estimated by MADNet 2.0 [12] from the original HiRISE patch are used as a geometric initialisation for the Gaussian point cloud, reducing the number of floater artefacts reported in structure-from-motion-initialised 3DGS on textureless terrain [16].

IV. DATASET DESIGN AND ACQUISITION STRATEGY

A. HiRISE Nadir Corpus

HiRISE images are downloaded from the NASA PDS Geosciences Node. The primary geographic focus is Oxia Planum (18.3° N, 335.5° E), the candidate ExoMars 2028 landing site, selected because it has been observed by HiRISE on more than 30 occasions between 2007 and 2024. Secondary patches are drawn from geologically similar mid-latitude clay-bearing terrain in the Meridiani Planum and Gale Crater regions to improve distributional coverage.

Each HiRISE product is radiometrically corrected (RDR product type) and projected to the Mars Equidistant Cylindrical coordinate system before tiling. Tiles of 512×512 pixels at 25 cm/pixel correspond to ground footprints of 128×128 m. Tiles with cloud or data-gap fill fractions exceeding 5% are discarded. The target nadir corpus size is 50,000 patches, derived from approximately 80 full HiRISE strips. Each patch is stored as a 16-bit PNG to preserve radiometric depth, then normalised to $[0, 1]$ at training time. Solar azimuth and elevation metadata are extracted from the PDS label (SOLAR_LONGITUDE and SUB_SOLAR_AZIMUTH keywords) and stored as a companion JSON file per patch.

B. Rover Perspective Corpus

Rover NAVCAM images are obtained from two sources. Curiosity NAVCAM images (grayscale, 1024×1024 , 8-bit) are downloaded from the NASA Mars Exploration Rover raw image archive; Perseverance NAVCAM images (RGB, 1280×960 , 8-bit) are obtained from the NASA Mars 2020 raw image archive. Images from both missions are filtered to remove images acquired at solar longitudes corresponding to dust-storm opacity ($\tau_{\text{dust}} > 1.5$, flagged by the Mars Climate Database). Images showing the rover chassis, instrument hardware, or operator-planted markers are removed by a Faster-RCNN detector trained on 500 manually labelled examples.

Perseverance RGB images are desaturated to grayscale to match the Curiosity data before computing FID, though both are retained for separate training runs. The target rover corpus size is 30,000 images after filtering. No geographic alignment with the HiRISE nadir corpus is performed or required; the UNSB framework explicitly handles this unpaired setting.

C. Dataset Splits and Augmentation

Both corpora are split 80/10/10 into training, validation, and test subsets, stratified by geographic region to avoid spatial autocorrelation leakage. All augmentation is applied only to training data and consists of random horizontal and vertical flips, random 90° rotations, and random brightness jitter in $[-0.05, +0.05]$ normalised units. Solar-metadata angles are transformed consistently with any spatial flip or rotation. No generative augmentation is used during training of the Stage 2 model, to prevent circular self-distillation artefacts.

Table I summarises the corpus statistics.

TABLE I
DATASET SUMMARY

Corpus	Size	Resolution	Source
HiRISE nadir patches	50,000	25 cm/px	NASA PDS HiRISE RDR
Rover NAVCAM images	30,000	≤ 1280 px	NASA raw archives
Total	80,000	—	—

V. TRAINING STRATEGY AND OPTIMISATION

A. Stage 1 Super-Resolution

The Stage 1 diffusion model is initialised from the DiffusionSat checkpoint [1] (pre-trained on Sentinel-2 and NAIP imagery) and fine-tuned on Mars orbital data using a cosine annealing schedule with $T_{\text{max}} = 100,000$ iterations, peak learning rate $\eta = 2 \times 10^{-5}$, and a linear warmup over the first 2,000 iterations. The batch size is 16 (image pairs of 128×128 crops). Fine-tuning is expected to require approximately 48 GPU-hours on a single A100-80GB. Gradient clipping at $\|\nabla\| \leq 1.0$ is applied throughout.

B. Stage 2 Neural Schrödinger Bridge

The UNSB is trained following the iterative proportional fitting procedure of [3]: alternating forward and backward half-bridge updates, each lasting 5,000 gradient steps, for a total of 10 IPFP outer iterations (100,000 steps total). The optimiser is AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4} , and base learning rate $\eta = 1 \times 10^{-4}$, decayed by a factor of 0.5 at IPFP iterations 5 and 8. The batch size is 8 image pairs (512×512); effective batch size is doubled by gradient accumulation over 2 steps.

The four auxiliary loss networks — the shadow-direction estimator D_ψ , the Hapke inversion network Ω_ξ , and the geological classifier K_κ — are pre-trained separately and kept frozen during UNSB training. D_ψ is a MobileNetV3-Small regressor trained on synthetic rendered images with known sun directions (3,000 training examples generated by the SPICE toolkit). Ω_ξ is a ResNet-18 trained on Hapke-forward-model renders parameterised by $\varpi_0 \in [0.12, 0.35]$. K_κ is a binary classifier (ResNet-50) trained to distinguish real HiRISE patches from out-of-distribution synthetic Mars images, using 5,000 positive (real) and 5,000 negative (Earth remote sensing) examples; it achieves 91% balanced accuracy on a held-out split.

The loss coefficients in Eq. 9 are initially set to $\lambda_1 = 0.5$, $\lambda_2 = 1.0$, $\lambda_3 = 0.8$, $\lambda_4 = 0.1$ and are tuned by random search over the validation set across 20 configurations. Estimated total Stage 2 training time is 72 GPU-hours on four A100-80GB cards with data-parallel training.

The pseudo-code for the full Stage 2 training loop is given in Algorithm 2.

C. Stage 3 Gaussian Splatting

Stage 3 uses the reference 3DGS implementation [5] without architectural modification. The initialisation point cloud ($\sim 50,000$ points) is seeded from MADNet-2.0 depth estimates [12] projected into 3D using the nadir camera model. Training

Algorithm 1: Stage 2 UNSB Training with Physics Losses

Input: HiRISE corpus $\mathcal{X}$; rover corpus $\mathcal{Y}$; pretrained auxiliary nets $D_\psi, \Omega_\xi, K_\kappa$; loss weights $\lambda_{1..4}$; IPFP iterations $N_{\text{ipfp}} = 10$

Output: Trained generator G_ϕ

```

for  $n = 1$  to  $N_{\text{ipfp}}$  do
  // Forward half-bridge: update  $G_\phi$ 
  for  $t = 1$  to 5000 do
    Sample  $\mathbf{x} \sim \mathcal{X}, \mathbf{y} \sim \mathcal{Y}$ 
     $\hat{\mathbf{y}} \leftarrow G_\phi(\mathbf{x})$ 
    Compute  $\mathcal{L}_{\text{SB}}, \mathcal{L}_{\text{solar}}, \mathcal{L}_{\text{photo}}, \mathcal{L}_{\text{geo}}, \mathcal{L}_{\text{perceptual}}$  (Eqs. 4–8)
     $\mathcal{L} \leftarrow \mathcal{L}_{\text{total}}$  (Eq. 9)
    Update  $G_\phi$  via AdamW( $\nabla_\phi \mathcal{L}$ )
  end for
  // Backward half-bridge: update backward net
  for  $t = 1$  to 5000 do
    Update backward drift following [3] IPFP step
  end for
end for
return  $G_\phi$ 

```

Fig. 2. Pseudocode for Stage 2 UNSB training. The outer loop iterates the Iterative Proportional Fitting Procedure (IPFP) for $N_{\text{ipfp}} = 10$ outer steps; each step alternates 5,000 forward-half-bridge gradient updates (training G_ϕ with the combined physics loss) and 5,000 backward-half-bridge updates. Auxiliary nets $D_\psi, \Omega_\xi, K_\kappa$ are frozen throughout.

runs for 30,000 iterations at a fixed learning rate of 5×10^{-4} for position parameters and 1.5×10^{-3} for colour parameters, with densification every 500 iterations and opacity pruning below 10^{-3} .

VI. EVALUATION FRAMEWORK

A. Perceptual Quality

The Fréchet Inception Distance (FID) [10] measures distributional similarity between generated images and held-out real rover images. FID is computed over 10,000 generated samples and 10,000 held-out NAVCAM images using InceptionV3 features. Lower FID indicates greater distributional fidelity; state-of-the-art unpaired Mars image translation has not been benchmarked on this exact corpus, so the proposed method is compared against the four baselines defined in Section VI-F. The Learned Perceptual Image Patch Similarity (LPIPS) metric is also reported.

B. Structural Fidelity Against Synthetic Ground Truth

For a subset of test patches where a high-quality Digital Terrain Model (DTM) is available — specifically 200 HiRISE stereo-derived DTMs from Oxia Planum — synthetic ground-truth perspective renders are generated using the Mesa 3D software rasteriser at known viewpoints. Structural Similarity Index (SSIM) and LPIPS are computed between generated and DTM-rendered images. This metric measures whether the geometry implied by the generated imagery is consistent with actual Mars topography, complementing the distributional FID measure.

C. Physical Plausibility Metrics

- *Shadow angle error:* Mean absolute angular error (degrees) between the dominant shadow direction predicted by D_ψ on generated images and the ground-truth solar azimuth derived from image metadata. Reported as the mean and standard deviation over the test set.
- *Hapke BRDF residual:* Root-mean-squared deviation of the estimated single-scattering albedo $\hat{\omega}_0$ from the Mars prior $\mathcal{N}(0.22, 0.05^2)$, reported in albedo units.

D. Hallucination Rate

The geological classifier K_κ (Section V) is applied to generated test patches. The hallucination rate is defined as the fraction of generated patches assigned the out-of-distribution label (i.e. those that do not pass the geological plausibility threshold). This metric directly quantifies the scientific unreliability concern raised by [10], [11].

E. VR Rendering Quality

Gaussian Splatting scenes constructed from Stage 3 outputs are benchmarked on render throughput (frames per second at 1920×1080 on an NVIDIA RTX 4090) and on novel view synthesis quality (PSNR, SSIM) measured against held-out viewpoints not used during 3DGS training. A user study ($n \geq 20$ participants) assessing perceptual realism on a 7-point Likert scale will be conducted if scheduling permits within the dissertation timeline.

F. Baseline Methods

Table II compares the proposed method against four baselines across five axes relevant to the task.

The MARS-GAN baseline [7] requires paired data from the same geographic location at two resolutions, which is unavailable for most of the Martian surface. DiffusionSat [1] fine-tuned on Mars performs nadir super-resolution and conditional inpainting but does not perform view-angle translation; it is included to isolate the contribution of the Schrödinger Bridge. CycleGAN provides an unpaired translation baseline with no physics grounding. The UNSB-without-physics-loss ablation directly isolates the contribution of $\mathcal{L}_{\text{solar}}, \mathcal{L}_{\text{photo}},$ and $\mathcal{L}_{\text{geo}}$ within the proposed framework.

VII. CONCLUSION

This chapter has presented the complete methodology for a three-stage physics-constrained generative pipeline that synthesises rover-perspective Mars surface imagery from orbital inputs and reconstructs the result as a real-time Gaussian Splatting VR scene. The key methodological contributions are: (i) the application of the Unpaired Neural Schrödinger Bridge to the Mars nadir-to-perspective domain gap, which is the first such application to planetary remote sensing; (ii) four physics-informed loss terms rooted in solar geometry, the Hapke BRDF [4], and Mars planetary science, which constrain the generated imagery to be physically plausible rather than merely perceptually realistic; and (iii) an evaluation protocol that goes beyond standard FID to measure shadow

TABLE II
COMPARISON OF PROPOSED METHOD AGAINST BASELINE METHODS

Method	Training data	View transformation	Physics constraints	Output format	Est. FID↓
MARSGAN [7]	Paired CTX–HiRISE (same site)	Nadir SR only	None	2D image	~180
DiffusionSat (Mars fine-tune) [1]	Unpaired nadir + conditioning text	Nadir SR / inpainting	None	2D image	~140
CycleGAN (naïve)	Unpaired nadir + rover	Nadir-to-perspective	None	2D image	~110
UNSB (no physics loss) [3]	Unpaired nadir + rover	Nadir-to-perspective	None	2D image	~80
Proposed (Ours)	Unpaired nadir + rover	Nadir-to-perspective 3D VR	Solar, Hapke, geological, perceptual	3DGS scene	≤55 (target)

↓ Lower is better. Estimated FID values for baselines are derived from analogous experiments in the cited literature adapted to Mars imagery; the proposed method target is based on the physics-constraint ablation analysis in [15] showing ~30% FID reduction from physics-informed losses.

angle error, Hapke BRDF residual, and geological hallucination rate, providing a quantitative link between generative model outputs and planetary science ground truth.

The pipeline targets an FID of ≤ 55 on held-out real NAVCAM imagery, a shadow-angle error below 10° , a hallucination rate below 5%, and VR renders at >72 fps. These targets represent substantial improvements over the four baseline methods and, if achieved, would constitute the first demonstrated system for realistic, physics-consistent Mars landing site visualisation from orbital imagery alone. The methodology is designed to be modular: Stage 2 and Stage 3 operate independently of Stage 1 wherever HiRISE data are already available, and the physics-loss framework is extensible to other planetary bodies for which Hapke-model parameters and sun-geometry metadata can be derived.

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